



Course syllabus

Faculty of Technology

Department of Mathematics

1MA406 Linjär algebra, 7,5 högskolepoäng

Linear Algebra, 7.5 credits

Main field of study

Mathematics

Subject Group

Mathematics

Level of classification

First Level

Progression

G1N

Date of Ratification

Approved by Faculty of Technology 2018-10-15

The course syllabus is valid from autumn semester 2019

Prerequisites

General entry requirements and Chemistry 1, Mathematics 4, Physics 2 or Chemistry A, Mathematics E, Physics B (Field-specific entry requirements 9/A9).

Objectives

After completing the course, the student should be able to

- solve problems, perform calculations, and conduct lines of reasoning within the part of mathematics that is covered by the course, and to communicate these solutions, calculations, and reasonings in writing
- describe definitions, and formulate and prove theorems that are central to the course.

Content

The main goal with the course is to give an introduction to linear algebra, its applications and the use of mathematical software. Contents:

- Linear systems of equations,
- Matrices, rank, determinants,
- Vector spaces, vectors, subspaces, basis and dimension, change of basis, row space and column space,
- Vector geometry: Inner product, vector product, volume and determinants. Lines and planes, angles- and distances,
- Linear transformations, matrix representation, kernel, image and range,
- Orthogonality: inner product in $\mathbb{R}^n$, orthogonal subspaces, the least square method,
- Diagonalization,
- Problemsolving using mathematical software, Matlab or the equivalent.

Theory and methods are illustrated by examples and problem solving from mechanics, electronics, economics/operations research and fitting curves.

Type of Instruction

Lectures and seminars.

Examination

The course is assessed with the grades A, B, C, D, E, Fx or F.

The grade A constitutes the highest grade on the scale and the remaining grades follow in descending order where the grade E is the lowest grade on the scale that will result in a pass. The grade F means that the student's performance is assessed as fail (i.e. received the grade F).

The student's knowledge in problem solving and theory is assessed in the form of written examination. Problemsolving using mathematical software is assessed in the form of assignments. The final grade is determined by a weighted average of the result of the two examinations.

Course Evaluation

During the course or in close connection to the course, a course evaluation is to be carried out. The result and analysis of the course evaluation are to be communicated to the students who have taken the course and to the students who are to participate in the course the next time it is offered. The course evaluation is carried out anonymously. The compiled report will be filed at the Faculty.

Credit Overlap

The course cannot be included in a degree along with the following courses of which the content fully, or partly, corresponds to the content of this course: 1MA403 Vector Geometry, 1MA133 Linear algebra for engineers.

Other

Grade criteria for the A–F scale are communicated to the student through a special document. The student is to be informed about the grade criteria for the course by the start of the course at the latest.

Required Reading and Additional Study Material

Steven Leon, *Linear Algebra with Applications*, Pearson, latest edition. Expected reading 275/485 pages.

Franco Vivaldi, *Mathematical Writing*, Springer, 2014. Expected reading 50/204 pages.

David Lay, *Linear Algebra and Its Applications*, Pearson, latest edition.

David Poole, *Linear Algebra: a modern introduction*, 4th edition or later, 2014.

Kenneth Hardy, *Linear Algebra: for Engineers and Scientists*, Pearson, 2006. Expected reading 323/480 pages.